

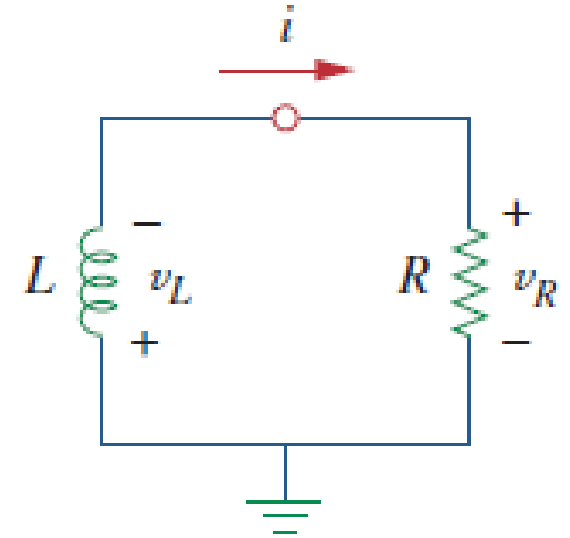
Module -2

Transient Response of First Order RL Circuits

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The Source Free RL circuit

- A source free RL circuit occurs when its dc source is suddenly disconnected.
- The energy already stored in the inductor is released to the resistors.
- To determine the current through the inductor is the objective.



- Assume that the inductor has an initial current, I_0

$$i(0) = I_0$$

- The corresponding energy stored in the inductor is

$$w(0) = \frac{1}{2} L I_0^2$$

- Applying KVL around the loop

$$v_L + v_R = 0$$

$$L \frac{di}{dt} + Ri = 0$$

$$\frac{di}{dt} + \frac{R}{L}i = 0$$

Rearranging terms and integrating gives

$$\int_{I_0}^{i(t)} \frac{di}{i} = - \int_0^t \frac{R}{L} dt$$

$$\ln i \Big|_{I_0}^{i(t)} = - \frac{Rt}{L} \Big|_0^t$$

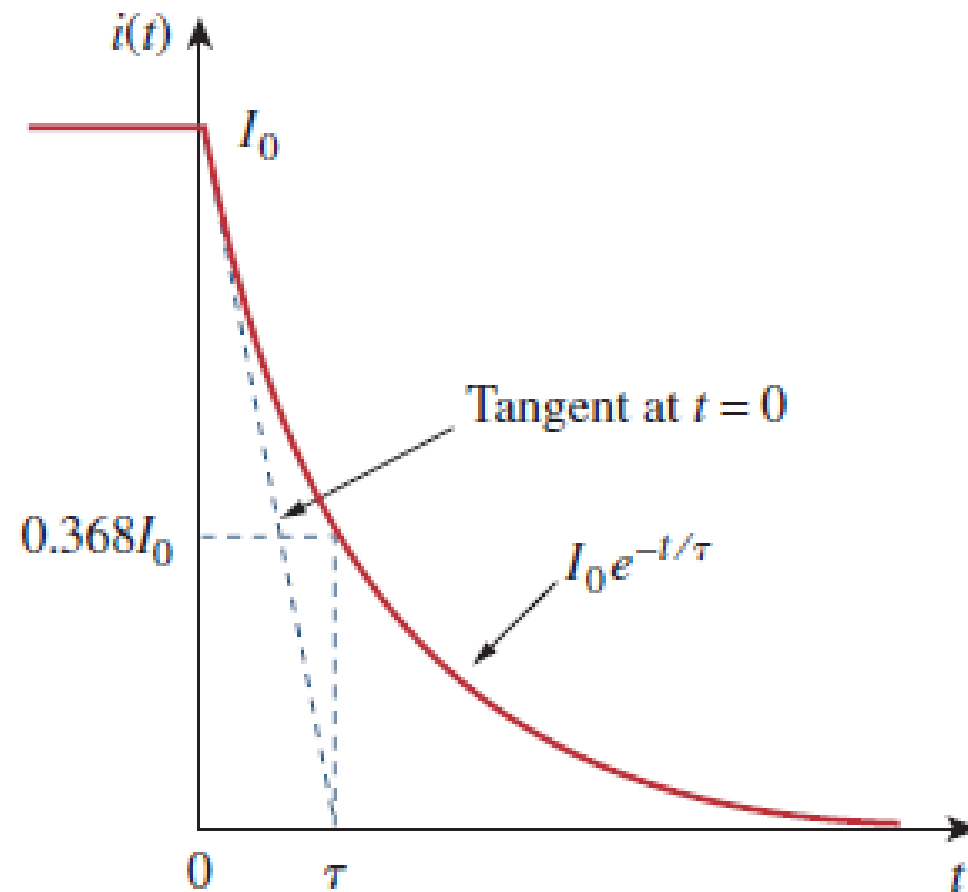
$$\ln i(t) - \ln I_0 = -\frac{Rt}{L} + 0$$

$$\ln \frac{i(t)}{I_0} = -\frac{Rt}{L}$$

Taking the powers of e ,

$$i(t) = I_0 e^{-Rt/L}$$

The current response of the RL circuit.



$$\tau = \frac{L}{R}$$

$$i(t) = I_0 e^{-t/\tau}$$

$$v_R(t) = iR = I_0 R e^{-t/\tau}$$

The power dissipated in the resistor is

$$p = v_R i = I_0^2 R e^{-2t/\tau}$$

The energy absorbed by the resistor is

$$w_R(t) = \int_0^t p(\lambda) d\lambda = \int_0^t I_0^2 e^{-2\lambda/\tau} d\lambda = -\frac{\tau}{2} I_0^2 R e^{-2\lambda/\tau} \Big|_0^t,$$

$$w_R(t) = \frac{1}{2} L I_0^2 (1 - e^{-2t/\tau})$$

$$t \rightarrow \infty, w_R(\infty) \rightarrow \frac{1}{2} L I_0^2,$$

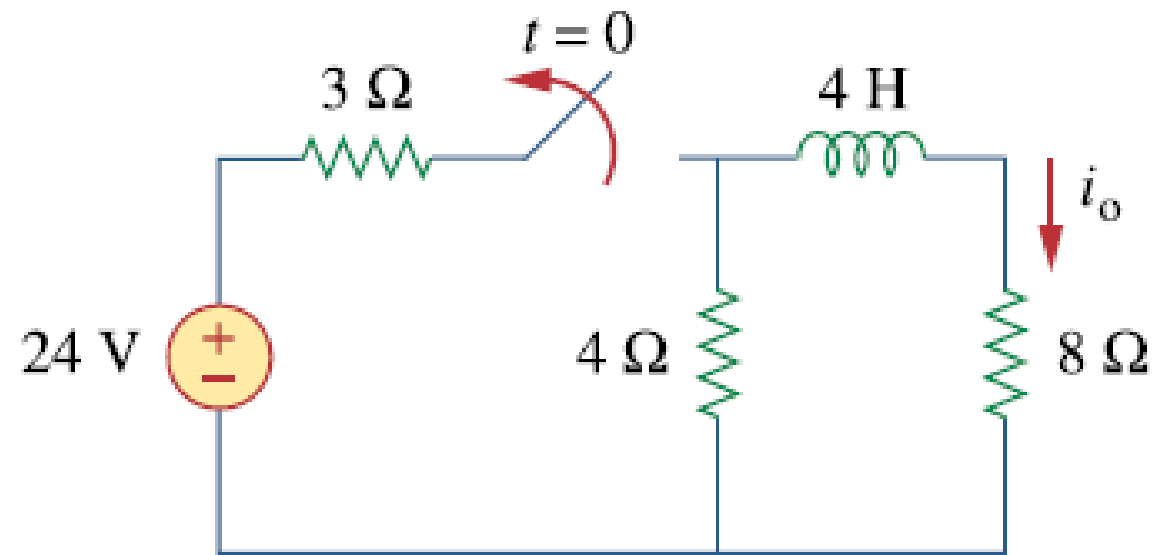
- **Steps to solve the Source free RL circuit:**

1. Find the initial current I_0 through the inductor.
2. Find the time constant $\tau = \frac{L}{R}$
3. Find the current through the inductor.

$$i(t) = I_0 e^{-t/\tau}$$

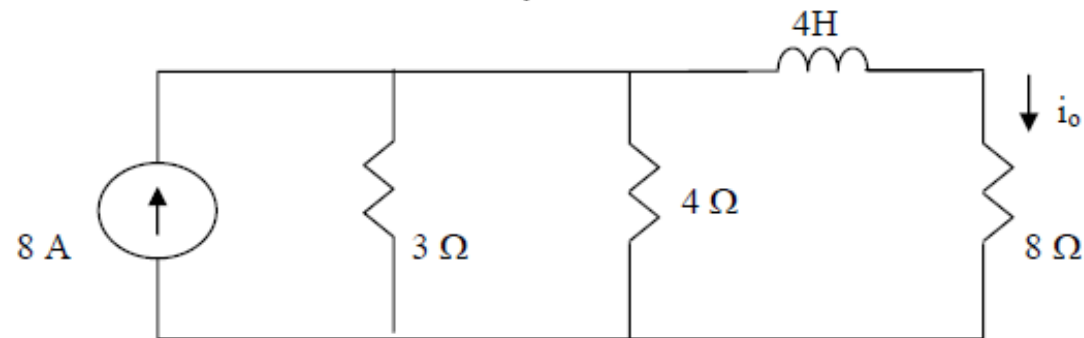
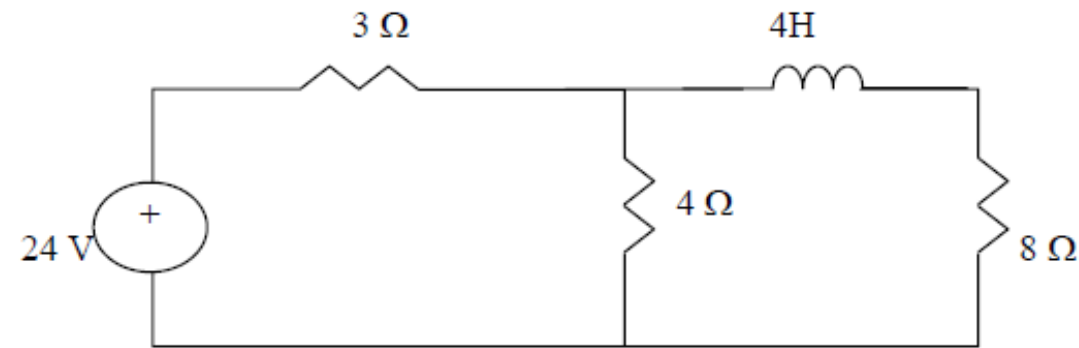
Numerical -1

Find i_o for $t > 0$.



- Solution:

For $t < 0$, we have the circuit shown below.



$$3 // 4 = \frac{4 \times 3}{4 + 3} = 1.7143$$

$$i_o(0^-) = \frac{1.7143}{1.7143 + 8} (24) = 1.4118 \text{ A}$$

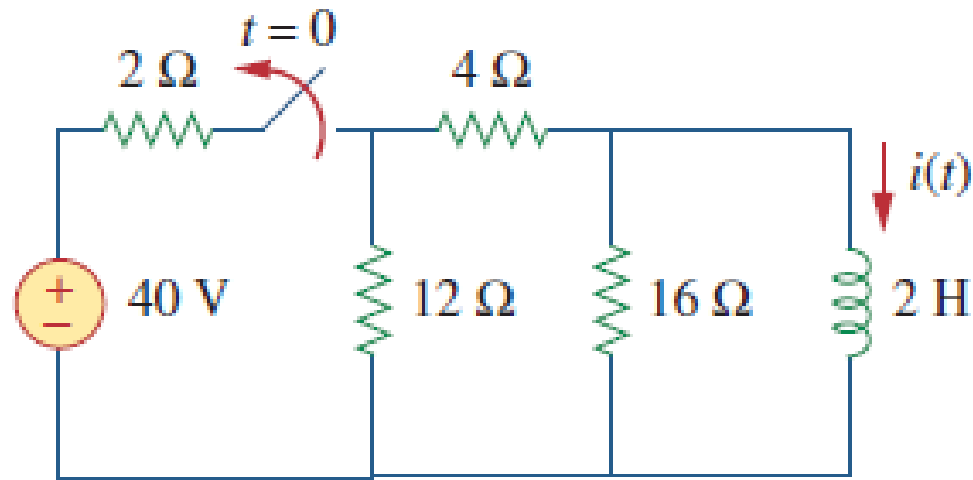
For $t > 0$, we have a source-free RL circuit.

$$\tau = \frac{L}{R} = \frac{4}{4 + 8} = 1/3$$

$$i_o(t) = i_o(0) e^{-t/\tau} = \underline{1.4118 e^{-3t} \text{ A}}$$

Numerical -2

The switch in the circuit has been closed for a long time. At time $t=0$, the switch is opened. Calculate $i(t)$ for $t>0$.



Solution:

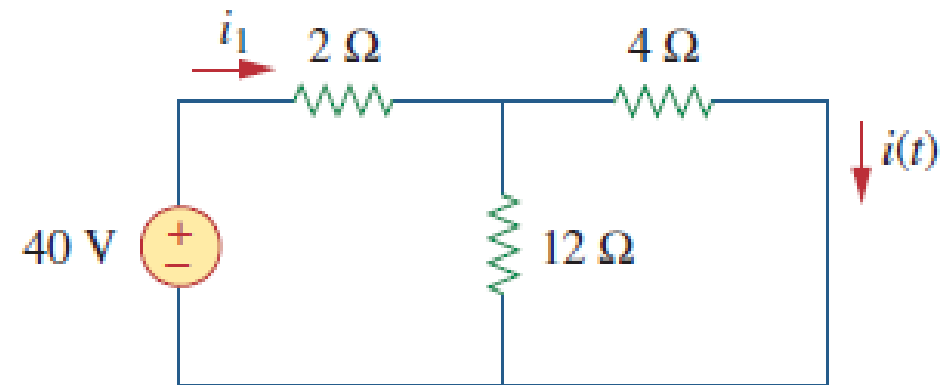
When $t < 0$, the switch is closed, and the inductor acts as a short circuit to dc. The $16\text{-}\Omega$ resistor is short-circuited;

$$\frac{4 \times 12}{4 + 12} = 3\ \Omega$$

$$i_1 = \frac{40}{2 + 3} = 8\text{ A}$$

$$i(t) = \frac{12}{12 + 4} i_1 = 6\text{ A}, \quad t < 0$$

$$i(0) = i(0^-) = 6\text{ A}$$



When $t > 0$, the switch is open and the voltage source is disconnected.

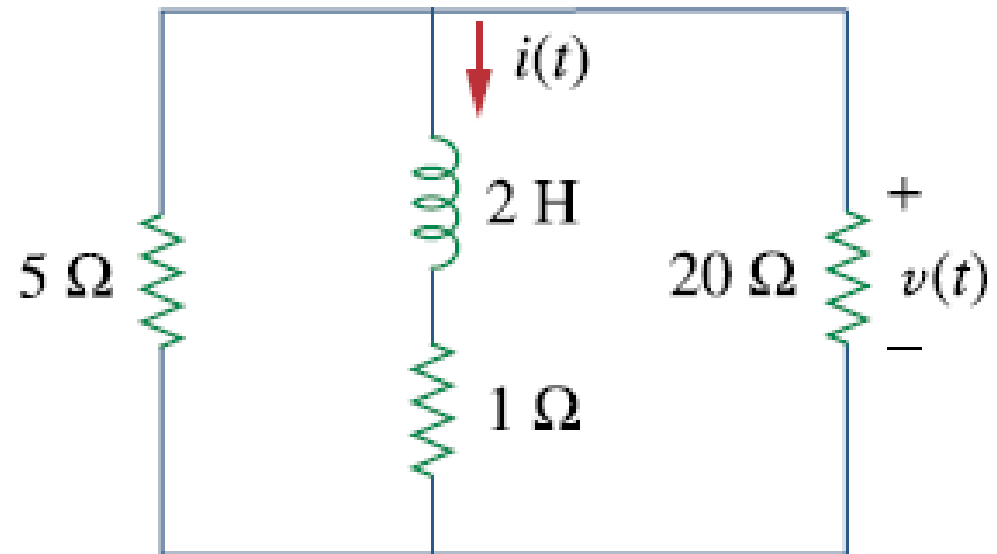
$$R_{\text{eq}} = (12 + 4) \parallel 16 = 8 \, \Omega$$

$$\tau = \frac{L}{R_{\text{eq}}} = \frac{2}{8} = \frac{1}{4} \text{ s}$$

$$i(t) = i(0)e^{-t/\tau} = 6e^{-4t} \text{ A}$$

Numerical -3

Find $i(t)$ and $v(t)$ for $t > 0$ in the circuit . Assume $i(0) = 10$ A



- Solution:

$$i(t) = i(0) e^{-t/\tau}, \quad \tau = \frac{L}{R_{eq}}$$

$$R_{eq} = 5 \parallel 20 + 1 = 5 \, \Omega, \quad \tau = \frac{2}{5}$$

$$i(t) = \underline{\mathbf{10 e^{-2.5t} \, A}}$$

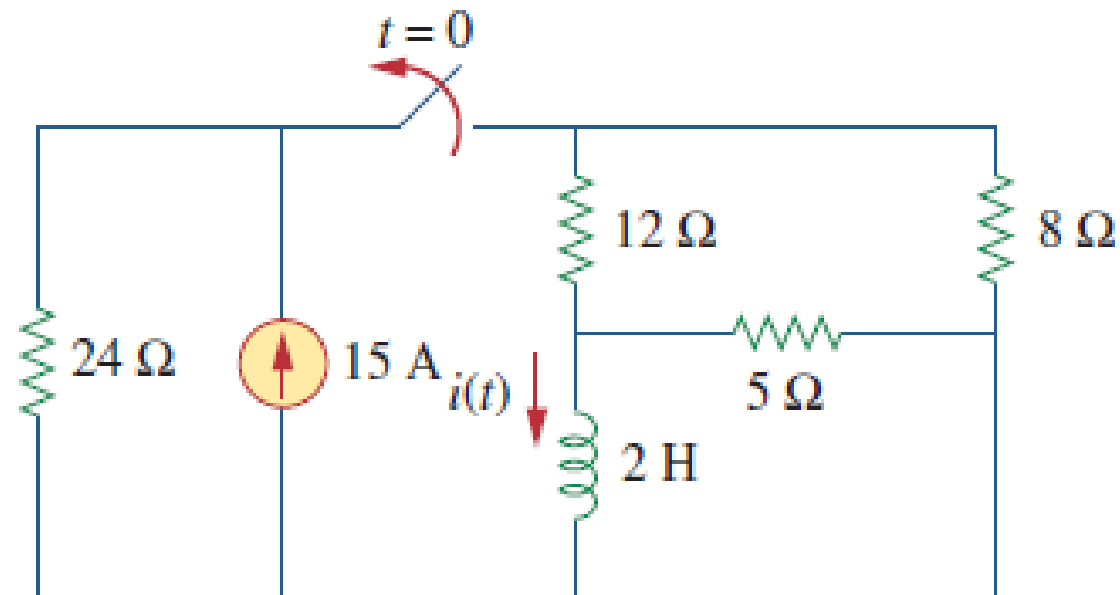
Using current division, the current through the 20 ohm resistor is

$$i_o = \frac{5}{5+20} (-i) = \frac{-i}{5} = -2 e^{-2.5t}$$

$$v(t) = 20 i_o = \underline{\mathbf{-40 e^{-2.5t} \, V}}$$

Practice problem-1

find $i(t)$ for $t > 0$.



Answer: $5e^{-2t}\text{ A}, t > 0$.